

Power BI Stock Market Dashboard Price and Volume Trend Analysis

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1. Introduction

This report looks at a stock market dashboard that I built in Power BI. The dashboard uses daily data from May 2019 to December 2025 and brings price and trading volume together on one page, so that the two can be studied side by side instead of in separate sheets.

The page has two charts, three filter controls and one summary table. The charts show how volume and closing price behave over time. The filters let the reader change the time period or the price type without editing anything. The summary table gives the exact numbers behind what is on screen. Each of these is explained in the sections below, followed by a short conclusion.

2. Objective

The aim of the dashboard is to answer a simple question: does heavy trading activity line up with big moves in price? To look into this, the report uses:

- a chart that compares the 20-day and 50-day moving averages of volume with the closing price,
- a chart that compares the raw daily volume with the 20-day moving average,
- filters for date range and price type, and
- a summary table with the key figures.

3. Visual 1: Volume 50 SMA, Volume 20 SMA and Close Price



Figure 1: Volume 50 SMA, Volume 20 SMA and Close Price by Date

This is a combination chart. The two volume moving averages are drawn as stacked areas and use the left axis, which is in millions of shares. The orange line is the daily closing price and uses the right axis. The date runs along the bottom from 2019 to 2025.

What stands out

- Volume is highest around early 2020 and again in mid 2022, when the shaded areas rise well above their usual level, reaching a little over 100 million at the peak.
- The closing price climbs from a low level in 2019 to a peak in late 2021 or early 2022, then falls sharply through 2022. It stays low for a while and starts recovering in 2024 and 2025.
- The heavy volume in 2022 sits right in the middle of that fall, which suggests a lot of selling while the price was dropping.
- Because the two volume areas are stacked, the top edge shows the two averages added together. To compare them properly, each area has to be read from its own baseline.

4. Visual 2: Volume 20 SMA and Total Volume

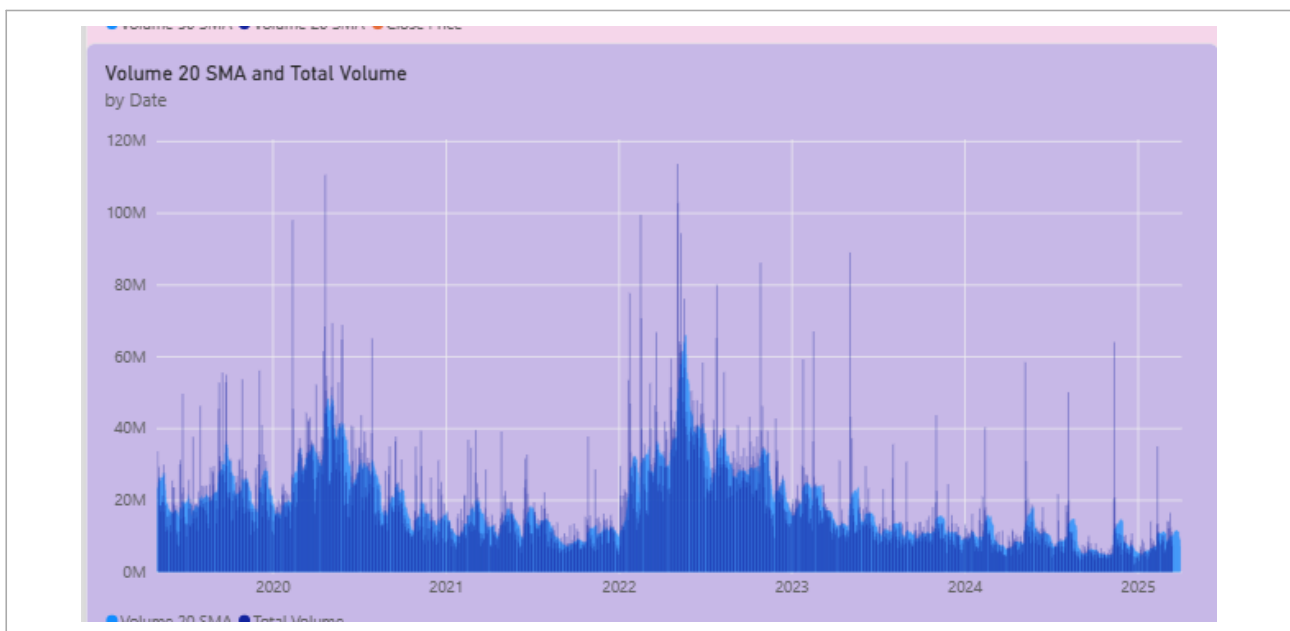


Figure 2: Volume 20 SMA and Total Volume by Date

The second chart plots the actual volume traded on each day as thin vertical bars, with the 20-day moving average as a smoother band underneath. The moving average takes out the day to day noise, while the bars show exactly which days were unusually busy.

What stands out

- Clusters of high volume appear around 2020 and 2022, which matches Figure 1.
- Some single days go far above the average. A few reach close to 100 million shares, and these are easy to miss in a smoothed chart.
- From 2023 onwards the average settles at a much lower level, with only occasional spikes, and it stays low into 2025.
- Read together, the two charts show both the general trend in activity and the individual days that caused it.

5. Filters and Controls

Three controls sit on the report page. Since all the visuals use the same Date field, changing a filter updates the whole page at once.

5.1 Date range slider

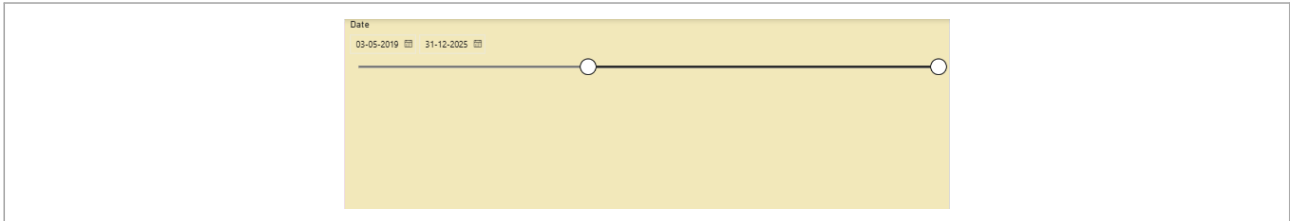


Figure 3: Date range slider

This is a between-type slider, currently set from 03-05-2019 to 31-12-2025. Either handle can be dragged to shorten the period. It is useful for questions like what happened during 2020, or what the last six months looked like.

5.2 Price metric slicer



Figure 4: Price metric slicer

This slicer lists four options: Close Price, High Price, Low Price and Open Price. In the screenshot only High Price is ticked. It lets the reader look at the data using a different price measure, for example the day's high instead of the close, without changing the report itself.

5.3 Relative date filter

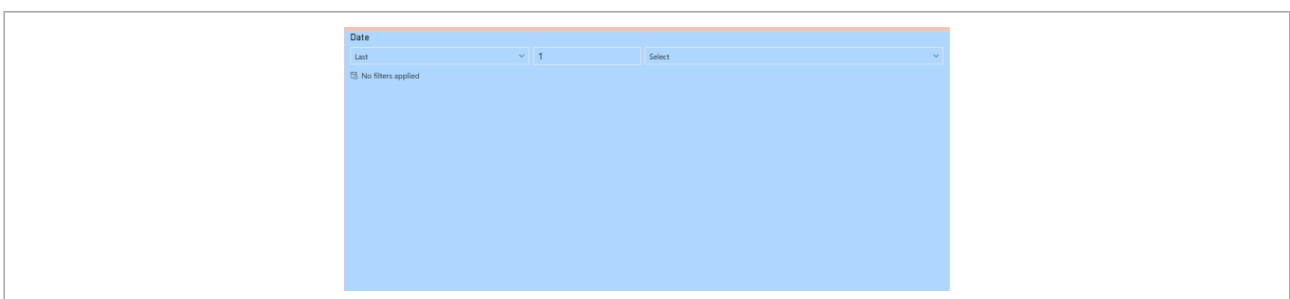


Figure 5: Relative date filter

This filter has three parts: a condition ("Last"), a number (currently 1) and a unit dropdown that is still on "Select". Once a unit such as days, months or years is chosen, the report shows a rolling window like the last 1 year. The message "No filters applied" means it is switched off right now, so the full date range is showing. It is quicker than the slider when an exact period is needed.

6. Summary Table



Figure 6: Summary table (Close Price, High Price, Open Price, Total Volume, Daily)

The last visual is a small table that lists the Close Price, High Price, Open Price, Total Volume and a Daily column. The values shown are:

Close Price	High Price	Open Price	Total Volume
48.46	176.29	48.46	40,912,544,200

The total volume of about 40.9 billion shares is far larger than any single day on the charts, where the busiest days are around 100 million. So these numbers are most likely totals or aggregates for the whole selected date range and not one day's figures. This also explains why the High Price (176.29) is so much higher than the Open and Close (both 48.46), since a high over several years will naturally be well above a single price value. The scrollbar under the table shows there are more columns that are cut off in the screenshot.

The table changes with the filters above, so it works as a quick numeric check on whatever period is selected.

7. Conclusion

The dashboard puts price and volume in one place and makes it easy to see how they relate. The two charts show that volume tends to rise sharply during big price moves, most clearly in 2020 and 2022, and the raw volume chart shows individual heavy days that the moving averages smooth out. The date slider, the relative date filter and the price metric slicer let the reader change the period or the price type in a few clicks, and the summary table gives the exact numbers for the selection.

One limitation is that the report only shows a relationship between volume and price. It does not say what caused any particular move. The dashboard could be improved by adding a Low Price line, labelling the summary table as totals or single-day values, and setting the relative date filter to a default unit. Even so, it does what it was built for, which is to give a clear view of how this stock has traded from 2019 to 2025.